Exact Synthesis of Two-Qubit Clifford+CS Circuits with a Near-Minimal Number of K and CS Gates

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We give a method for the decomposition of an arbitrary two-qubit Clifford+CS operator (equivalently a unitary with entries in $\mathbb{D}[i] = \mathbb{Z}[\frac{1}{2}, i]$, the Gaussian integers with dyadic denominators), into a Clifford+CS circuit. Given unitary U, let k be the least natural number such that entries of $(1+i)^k U$ are all Gaussian integers, then this method decomposes U, upto a global phase, into a circuit of length at most $20k + 9$, using at most $2k$ scaled H gates and at most $2k + 1$ Controlled-S gates, without using any ancilla. Moreover, the number of scaled H gate is at most 2 times worse than optimal, the number of *CS* gate is at most 4 times worse than optimal. This method is similar to the work of Russell [16] for the Clifford+T gate set but with more detailed analysis, where the ring in use is $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$ which contains $\mathbb{Z}[\frac{1}{2}, i]$.

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I. INTRODUCTION

In quantum computation, the decomposition of unitary operators into gates from a fixed universal set is an important problem. In this paper we focus on the exact synthesis 2-qubit Clifford+Controlled-S. Previous work on synthesis of Clifford+CS operators or Clifford+T operators mainly focus on minimizing the use of CS gate or T gate since they are more expensive than Clifford operators in certain scenerio. But works on T gate optimization [1, 3, 6, 7, 11, 13] suggested that the H gate is an "obstruction" to their T-optimization method. Quoting from [1] summarizes the reason: "It has been shown that T-depth optimizations can be implemented efficiently for circuits consisting only of T and CNOT gates and that H gates impede the optimization significantly".

We propose a method that produces circuits with near-minimal number of both the scaled H gate and CS gate, hence alleviate such "obstruction".

A. Background

Consider the following unitary operators:

$$i, \quad K = e^{-i\pi/4} H = \frac{1}{1+i} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix},$$

$$CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

Here, i is a scalar, namely the usual complex unit. H is the Hadamard gate, and K is a scaled version of the Hadamard gate. S is sometimes called the *phase gate*, and CZ is the controlled-Z gate. When closed under multiplication, identities, and tensor products, these operators generate the *Clifford group* (possibly up to scalars, depending which scalars are included in the Clifford group — for our purposes, the scalars ± 1 and $\pm i$ are sufficient).

Every operator U obtained in this way is of size $2^n \times 2^n$ for some natural number n, and as usual, we say that U is an operator on n qubits. We write $\mathcal{CLI}(n)$ for the group of n-qubit Clifford operators (just $\mathcal{CLI}$ if $n = 2$). It is well-known that this group is finite for any given n (see, e.g., [17]), and therefore not universal for quantum computing. We can obtain a universal gate set by adding the controlled phase gate

$$CS = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & i \end{bmatrix}.$$

The resulting operators are called the Clifford+CS operators. Note that $CZ = CS^2$. The set of 2-qubit Clifford+CS operators is denoted by $\mathcal{CCS}$. The set of 2-qubit Clifford+CS circuits (a circuit is a finite sequence of gates) is denoted by $\mathcal{CIR}$. Each circuit $D \in \mathcal{CIR}$ implements a unique operator $[D] \in \mathcal{CCS}$.

We use the following notations: we write $K_0 = K \otimes I, K_1 = I \otimes K$, and similarly for S_0 and S_1 where I is the 2×2 identity operator. We identify the scalar i with the 4×4-matrix $i\,(I \otimes I)$. In the notation for controlled gates, we use the convention that the target is the last index. For example, CX_{01} is a controlled X-gate with control qubit 0 and target qubit 1, also shorted as CX. CX_{10} is shorted as XC. The controlled S- and Z-gates are symmetric, in the sense that $CS_{jk} = CS_{kj}$ and $CZ_{jk} = CZ_{kj}$, and therefore the order of indices does not matter for them, so we just write CS and CZ. We also use the Z-gate, the X-gate, the controlled K-gate $CK = I \oplus K$, and the swap gate, denoted by Ex. They are related to the "primitive" gates introduced so far by $Z = S^2$, $X = KZKi$, $CX = K_1 CZ K_1 \cdot i$, $Ex = CX\,XC\,CX$, and

$$CK = CZ \cdot Z_0 S_0 Z_1 S_1 K_1 CS K_1 S_1 \cdot i. \tag{1}$$

Similar to CX gate, CK_{01} is shorted as CK and CK_{10} is shorted as KC. By Equation 1, $CK \in \mathcal{CCS}$.

B. Related work

There are many works on quantum circuit synthsis. Here we only list a few of the close related ones. Giles and Selinger first presented an algorithm for the exact synthesis of n-qubit operators with entries in the ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$ using a method involving decomposition into 1- and 2-level matrices, in which one ancilla is sufficient [2]. If the matrix entries are of the form $\frac{a+b\sqrt{2}+i(c+d\sqrt{2})}{\sqrt{2}^k}$, then this algorithm achieves an upper bound of $O(3^{2^n} nk)$ gates, which is far from optimal.

The problem with this algorithm was that the exponent in the denominator of the remaining entries of the matrix may increase after applying the 1- and 2-level matrices, and this increases the number of operations needed to reduce the matrix. Kliuchnikov then introduced an algorithm which achieved an upper bound of $O(4^n nk)$ gates, using a different method requiring at most two ancillas [3].

n	# T gates	Time complexity	# ancilla	Ref.
1	$\mathcal{T}(U)$	$O(\mathcal{T}(U))$	0	[7]
2	$\leqslant 10\mathcal{T}(U)$	$O(\mathcal{T}(U))$	0	this paper
$\geqslant 1$	$O\left(3^{2^n} nk\right)$	$O\left(\text{poly}\left(2^n, 3^{2^n} nk\right)\right)$	1	[15]
$\geqslant 1$	$O\left(4^n nk\right)$	$O\left(\text{poly}\left(2^n, 4^n nk\right)\right)$	2	[11]
$\geqslant 1$	$\mathcal{T}(U)$	$O\left(2^{n\mathcal{T}(U)/2}\right)$	0	[7,16]

TABLE I. Summary of exact synthesis algorithms. Here n is the number of qubits, $\mathcal{T}(U)$ is the T count, and k is the denominator exponent of the input unitary.

With the revelation that this efficient bound was possible, this paper presents an algorithm that achieves an efficient bound for the 1- and 2-qubit cases while using the original method of decomposition into 1- and 2-level operators in which one ancilla is sufficient.

II. SOME ALGEBRA

The concepts of this section are similar to those used in [2, 8–10, 14, 15], except in some cases the underlying ring is different.

A. Rings

As usual, we write $\mathbb{N} = \{0, 1, \ldots\}$ for the set of natural numbers. We also write $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$ to denote the rings of integers, real numbers, and complex numbers, respectively. We write $z^\dagger$ for the complex conjugate of a complex number. Recall that in any ring, a *unit* is an invertible element.

Let $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ be the ring of Gaussian integers. It is a Euclidean domain[4], and therefore, division with remainder, greatest common divisors, divisibility, and the concept of a prime (i.e. a non-unit that cannot be written as the product of two non-units) all can be used in $\mathbb{Z}[i]$. Let $\gamma = 1 + i$. It is well-known that γ is a Gaussian prime [12]. Note that γ is a divisor of 2, and γ^2 and 2 divide each other. Using Euclidean division, it is further easy to check that

$$\mathbb{Z}[i]/(\gamma) = \{0, 1\}, \quad \mathbb{Z}[i]/(2) = \mathbb{Z}[i]/(\gamma^2) = \{0, 1, i, 1+i\},$$

$$\text{and} \quad \mathbb{Z}[i]/(\gamma^3) = \{\pm 1, \pm i, 0, 1+i, 1-i, 2\}.$$

Note that the above is a slight abuse of notation. Normally, $\mathbb{Z}[i]/(\gamma)$ denotes the quotient ring of $\mathbb{Z}[i]$ modulo the ideal (γ) whose elements are residues nomrally denoted by $x \bmod (\gamma)$ for any $x \in \mathbb{Z}[i]$. Here we omit the mod (γ) for simlicity. Note that $\{0, 1\}$ etc. are complete

sets of residue classes and also the reminders from the Euclidean division of x divided by γ.

Let $\mathbb{D} = \mathbb{Z}[\frac{1}{2}] = \left\{\frac{a}{2^k} \mid a \in \mathbb{Z}, k \in \mathbb{N}\right\}$ be the ring of *dyadic fractions*, i.e., fractions whose denominator is a power of 2. Consider $\mathbb{D}[i] = \{r + si \mid r, s \in \mathbb{D}\} = \mathbb{Z}[\frac{1}{2}, i]$. For every $t \in \mathbb{D}[i]$, there exists some natural number k such that $\gamma^k t \in \mathbb{Z}[i]$. This motivates the following definition.

Definition II.1. Let $t \in \mathbb{D}[i]$. A natural number $k \in \mathbb{N}$ is called a *denominator exponent* for t if $\gamma^k t \in \mathbb{Z}[i]$. The least such k is called the *least denominator exponent* of t, denoted by $\mathrm{lde}_\gamma(t)$. Equivalently, the least denominator exponent of t is the smallest $k \in \mathbb{N}$ such that t can be written in the form $\frac{s}{\gamma^k}$, where $s \in \mathbb{Z}[i]$.

More generally, we say that k is a denominator exponent for a vector or matrix if it is a denominator exponent for all of its entries. The least denominator exponent for a vector or matrix is therefore the least k that is a denominator exponent for all of its entries.

Remark II.2. By enumerating all 2-qubit Clifford operators, we have that the lde of a 2-qubit Clifford operator is at most 2.

B. Residue ring

It is well-known that

$$\mathbb{Z}[i]/(\gamma^n) = \left\{\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n) \,|\, b_i \in \{0,1\}\right\}.$$

We give an elmentary proof by induction on n. Note that similar proof is used for proving the case for $\mathbb{Z}[\sqrt{2}]/(\sqrt{2}^n)$ by Clement[5]. When $n = 1$, the claim obviously holds. Suppose this is true for n, i.e., for any $x \in \mathbb{Z}[i]$, x is congruent to some $\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n)$, that is

$$x = \sum_{i=0}^{n-1} b_i\gamma^i + k\gamma^n \text{ for some } k.$$

Since $k = b_n + k'\gamma$ for some $b_n \in \{0,1\}$, substituting k in and some rewrite gives

$$x = \sum_{i=0}^{n} b_i\gamma^i + k'\gamma^{n+1}, \text{ i.e., } x \equiv \sum_{i=0}^{n} b_i\gamma^i \bmod (\gamma^{n+1}).$$

For simplicity, we omit the $\bmod (\gamma^n)$ in $\sum_{i=0}^{n-1} b_i\gamma^i \bmod (\gamma^n)$. Since $b_i \in \{0,1\}$, we write binary string $(b_0b_1...)_2$ for the summation, and the parenthesis and subscript $_2$ can be omitted if we can infer that $b_0b_1...$ are residues in binary format from the context.

Moreover, residue classes are distinct, i.e. if $b_0b_1...b_{n-1}$ and $b'_0b'_1...b'_{n-1}$ are equivalent, i.e., $\sum_{i=1}^{n-1} b_i\gamma^i - \sum_{i=1}^{n-1} b'_i\gamma^i = k\gamma^n$ for some k, then $b_i = b'_i$. Modding (γ) on both side gives $b_0 = b'_0$. Then cancelling b_0 and b'_i, and dividing both side by γ give

$$\sum_{i=1}^{n-1} b_i\gamma^{i-1} - \sum_{i=1}^{n-1} b'_i\gamma^{i-1} = k\gamma^{n-1}$$

Then modding (γ) on both side agian gives $b_1 = b'_1$. Repeat this process proves $b_i = b'_i$.

We write $\rho_n(x)$ for the binary residue $x \pmod{\gamma^n}$. From the proof, we see $\rho_n(x)$ is "stable" w.r.t increase of n, i.e., the first n digits of $\rho_{n+1}(x)$ is $\rho_n(x)$. Also ρ_2 is "stable" under taking complex conjugation, i.e., $\rho_2(x^\dagger) = \rho_2(x)$.

We sometimes also write $x \equiv_{\gamma^n} y$ as an abbreviation for $x \equiv y \bmod \gamma^n$. Modulo γ^3, we have

x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$
1	100	-1	101	i	111	$-i$	110
0	000	$1+i$	010	$1-i$	011	2	001

The $\pmod{\gamma^n}$ residue classes form a commutative ring and there is a canonical ring homomorphism from $\mathbb{Z}[i]$. In particular, modulo γ^2, the addition of the ring coincides with the bitwise exclusive or (denoted by $\oplus$) operation; $(10)_2$ is the ring identity; $(00)_2$ the zero of the ring; multiplication by $(01)_2$ is the right shift operation; by distributivity $11 \cdot b_0b_1 = 10 \cdot b_0b_1 + 01 \cdot b_0b_1 = b_0(b_1 \oplus b_0)$. Note that since $\rho_2(i) = 11$, if $\rho_1(x) = 1$, then either $\rho_2(x) = 10$ or $\rho_2(ix) = 10$. Similarly, modulo γ^3, the addition is

$$b_0b_1b_2 + b'_0b'_1b'_2 = (b_0 \oplus b'_0)(b_1 \oplus b'_1)(b_2 \oplus b'_2 \oplus (b_0 \cdot b'_0)),$$

where the $\cdot$ is the multiplication of the ring $\mathbb{Z}[i]/(\gamma)$, equivalently the "and" of two bits. The ring identity is 100, zero is 000. Multiplication of 010 is a right shit by 1 digit and multiplication of 001 is a right shit by 2 digits. The rest of the multiplication table can be obtained using distributativity. The additive inverse is

$$-(b_0b_1b_2) = b_0b_1(b_2 \oplus b_0).$$

By looking at the ρ_3 table, we see ρ_3 is not "stable" under taking complex conjugation, but instead, if $\rho_3(x) = abc$, then $\rho_3(x^\dagger) = ab\bar{c}$. Since $\rho_3(i) = 111$, if $\rho_1(x) = 1$, then there exists $k \in \{0,1,2,3\}$ such that $\rho_3(i^k x) = 100$.

Definition II.3. For $x \in \mathbb{Z}[i]$, if $\rho_1(x) = 0$, we say x is *even*, otherwise *odd*.

For example, $2 + 3i$ and $5 + 2i$ are odd, and $2 + 4i$ and $1 + 3i$ are even. In particular, in our context, by an even Gaussian integer we mean one that is divisible by γ, not necessarily by 2. However if $x \in \mathbb{Z}$, then x is even (odd) as an integer if and only if x is even (odd) as a Gaussian integer. Therefore, our definition of parity extends the usual one on the integers. The following lemmas are straightforward.

Lemma II.4. *Let $\alpha = a + bi \in \mathbb{Z}[i]$. Then α is odd if and only if $\|\alpha\|^2 = a^2 + b^2$ is an odd integer.*

Lemma II.5. *If $\alpha \in \mathbb{Z}[i]$ is odd, then $\alpha \equiv \pm 1, \pm i \pmod{\gamma^3}$. In other words, $\alpha \equiv i^e \pmod{\gamma^3}$ for some $e \in \{0, 1, 2, 3\}$. Moreover, $\alpha \equiv i^e \pmod{\gamma^2}$ for some $e \in \{0, 1\}$.*

Lemma II.6. *If $\alpha, \beta \in \mathbb{Z}[i]$ satisfy $\alpha\gamma \equiv_{\gamma^{n+1}} \beta\gamma$, then $\alpha \equiv_{\gamma^n} \beta$.*

Let k be a denominator exponent (not necessarily the least) for $x \in \mathbb{D}[i]$, we define the residue function $\rho_n^k(x) = \rho_n(\gamma^k x)$. We call $\rho_n^k(x)$ the (n, k) residue of x. If k is a denominator exponent for a vector or matrix A, $\rho_n^k(A)$ is defined entry-wise.

Lemma II.7. *For $t \in \mathbb{D}[i]$, if $k = lde_\gamma(t) > 0$, then $\gamma^k t$ is odd.*

Lemma II.8. *For any $x \in \mathbb{D}[i]$ with denominator exponent k, $\rho_2^k(\bar{x}) = \rho_2^k(x)$, where $\bar{\cdot}$ is the complex conjugation.*

Lemma II.9. *For A a matrix with lde k, B another matrix of proper size with lde 0, then $\rho_n^k(AB) = \rho_n^k(A)\rho_n^0(B)$ and $\rho_n^k(BA) = \rho_n^0(B)\rho_n^k(A)$.*

This is because $\rho_n^k(\cdot)$ is a ring homomorphism and matrix multiplication only involves ring operations on matrix entries.

C. Action on residues

Suppose $\rho_2(x) = b_0 b_1$, then $\rho_2(ix) = b_0(b_1 \oplus b_0)$. We say i acts on $b_0 b_1$ by sending it to $b_0(b_1 \oplus b_0)$.

Given an n residue $b_0 b_1 ... b_{n-1}$, a right shift of it is an $n+1$ residue $0 b_0 b_1 ... b_{n-1}$. Given an $n+1$ residue $0 b_0 b_1 ... b_{n-1}$, a left shift is $b_0 b_1 ... b_n$. Note that we choose to only allow "well-defined" shifts in the sense no information is lost and no unknown is introduced. Note that the usual right shift is used previously when explaining the multiplication of $\mathbb{Z}[i]/(\gamma^2)$ and $\mathbb{Z}[i]/(\gamma^3)$. We use $LS(x)$ for the left shift of x and RS for the right shift.

Lemma II.10. *For $t \in \mathbb{D}[i]$, k a denominator exponent of t, we have*

$$\rho_{n+1}^{k+1}(t) = \rho_{n+1}^k(\gamma t) = RS(\rho_n^k(t)) \text{ and}$$

and if $\gamma^k t$ is even,

$$\rho_n^{k-1}(t) = \rho_n^k(t/\gamma) = LS(\rho_{n+1}^k(t)).$$

Given a $x, y \in \mathbb{D}[i]$ with lde $k > 0$, we know $\rho_2^k(x) = 1b$ and $\rho_2^k(y) = 1b'$ for some $b, b' \in \{0, 1\}$. So $\rho_2^k(x+y) = 1b + 1b' = 0(b \oplus b')$. Then $\rho_1^{k-1}(x+y) = LS(\rho_2^k(x+y)) = b \oplus b'$. Similarly, $\rho_1^{k-1}(x-y) = LS(\rho_2^k(x-y)) = LS(1b - 1b') = LS((1-1)(b-b')) = b \oplus b'$ (subtraction in $\mathbb{Z}/(2)$ coincides with $\oplus$, in other words the additive inverse of b is itself). Note that if $b \oplus b' = 1$, then $\text{lde}(x \pm y)$ decreases by 1, and if $b \oplus b' = 0$, $\text{lde}(x \pm y)$ decreases by at least 2 if $k \geq 2$. Since dividing t by γ obviously increases $\text{lde}(t)$ by 1, we have for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 10b$ and $\rho_3^k(y) = 10b'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} 0(b \oplus b' \oplus 1) \\ 0(b \oplus b') \end{bmatrix}.$$

So we have $\text{lde}(K \begin{bmatrix} x \\ y \end{bmatrix}) = k - 1$ and one entry of it has lde $k-1$ and the other at most $k-2$ if $k > 2$.

In general for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 1b_0b_1$ and $\rho_3^k(y) = 1b_0'b_1'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} (b_0 \oplus b_0')(b_1 \oplus b_1' \oplus 1) \\ (b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 0b_0b_1$ and $\rho_3^k(y) = 0b_0'b_1'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} (b_0 \oplus b_0')(b_1 \oplus b_1') \\ (b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 1b_0b_1$ and $\rho_3^k(y) = 0b_0'b_1'$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b_0')(b_1 \oplus b_1') \\ 1(b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 0b_0b_1$ and $\rho_3^k(y) = 1b_0'b_1'$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b_0')(b_1 \oplus b_1') \\ 1(b_0 \oplus b_0')(b_1 \oplus b_1' \oplus 1) \end{bmatrix}.$$

III. SPECIAL UNITARIES

A. Permutation matrices

Let $a_0a_1a_2a_3$ be a permutation of $\{0, 1, 2, 3\}$. The permutation matrix $P_{a_0a_1a_2a_3}$ is the unitary matrix that maps e_i to e_{a_i}, where $\{e_i\}$ is the standard basis. All 24 permutation matrix are Clifford operators and they can be implemented by circuits of lenght at most 3, see Table II.

B. Diagonal unitary matrices

The only diagonal unitary matrices with entries in $\mathbb{D}[i]$ (and $\mathbb{Z}[i]$) are of the form

$$\text{diag}(i^a, i^b, i^c, i^d), \text{ where } a, b, c, d \in \{0, 1, 2, 3\}.$$

P	circuit	P	circuit	P	circuit
0123	identity	1023	$X_1 \cdot CX$	2103	$X_0 \cdot XC$
1203	$Ex \cdot X_0 \cdot XC$	2013	$Ex \cdot X_1 \cdot CX$	0213	Ex
3210	$X_1 \cdot X_0$	2310	$X_0 \cdot CX$	2130	$Ex \cdot X_1 \cdot XC$
3120	$Ex \cdot X_1 \cdot X_0$	1320	$Ex \cdot X_0 \cdot CX$	1230	$X_1 \cdot XC$
3012	$XC \cdot X_1$	0312	$Ex \cdot XC$	0132	CX
3102	$Ex \cdot CX \cdot X_0$	1302	$Ex \cdot X_0$	1032	X_1
3021	$Ex \cdot XC \cdot X_1$	0321	XC	0231	$Ex \cdot CX$
3201	$CX \cdot X_0$	2301	X_0	2031	$Ex \cdot X_1$

TABLE II. Circuit implementation of 24 permutation matrices.

They can all be implemented by Clifford+CS circuits with length of at most 6 upto a global phase i^k, using at most 1 CS gate. This because

$$Z_0^{b_0} Z_1^{b_1} S_0^{b_2} S_1^{b_3} CZ^{b_4} CS^{b_5} \cdot i^k, b_i \in \{0,1\}, k \in \{0,1,2,3\}.$$

are all distinct and diagonal, and it is of cardinality 4^4 which equals to the cardinality of the set of diagonal unitary matices.

C. Generalized unitary permutation matrices

A generalized unitary permutation matrix with entres in $\mathbb{D}[i]$ is a matrix with the same nonzero pattern as a permutation matrix (there is exactly one nonzero entry in each row and each column), but the nonzero entry can be any number of unit length, i.e, powers of i. It is easy to see all 4×4 generalized permutation matrices are product of a permutation matrix and a diagonal matrix, hence can be implemented by a circuit of lenght at most 9 containing at most 1 CS gate.

IV. THE EXACT SYNTHESIS ALGORITHM

A. Main theorems

Theorem IV.1. *For any $A \in U(\mathbb{D}[i])$ with lde k, there exists a circuit C implementing A, using at most $2k$ K gates, at most $2k+1$ CS gates, and at most $9(2k+1)+2k$ gates upto a global phase.*

Note that the proof (given later) is constructive, so it gives an algorithm for computing such circuit.

B. Optimality

Definition IV.2. Let D be a Clifford+CS circuit. Let $[D]$ be the unitary operator it implements. Define

- $\mathrm{rkc}(D) =$ the number of K gate that appears in D.
- $\mathrm{kc}(D) = \min\{\mathrm{rkc}(C) \mid C \in \mathcal{CCS} \text{ and } [C] = [D]\}$.
- $\mathrm{rcs}(D) =$ the number of CS gate that appears in D.
- $\mathrm{cs}(D) = \min\{\mathrm{rcs}(C) \mid C \in \mathcal{CCS} \text{ and } [C] = [D]\}$.
- $\mathrm{rlen}(D) =$ the number of gate that D contains.
- $\mathrm{len}(D) = \min\{\mathrm{rlen}(C) \mid C \in \mathcal{CCS} \text{ and } [C] = [D]\}$.

We say D is K-optimal, CS-optimal, and shortest if $\mathrm{rkc}(D) = \mathrm{kc}(D)$, $\mathrm{rcs}(D) = \mathrm{cs}(D)$, and $\mathrm{rlen}(D) = \mathrm{len}(D)$ repectively.

Any Clifford+CS circuit D is an alternating sequence of Clifford part C_i and a CS gate

$$D = C_0 \cdot CS \cdot C_1 \cdot CS \cdot \ldots \cdot C_n. \tag{2}$$

By enumerating all 2-qubit Clifford circuit, we know $\mathrm{lde}(C_i) \leq 2$. If (2) is CS-optimal, i.e., $\mathrm{cs}(D) = n$, then $\mathrm{lde}(D) \leq \mathrm{lde}(C_0) + \mathrm{lde}(CS) + \mathrm{lde}(C_1) \ldots \mathrm{lde}(C_n) \leq 2(n+1)$. We have the relation

$$\mathrm{lde}(D) \leq 2(\mathrm{cs}(D) + 1).$$

On the other hand, our synthesis algorithm output at most $2 \cdot \mathrm{lde}(D)$ K gates and $2 \cdot \mathrm{lde}(D) + 2$ CS gates. So our our synthesis algorithm is at most 4 times worse than optimal CS-count (experiments show it is 2 times worse, see Section **??**). To acheive the lde, we need at least $\mathrm{lde}(D)$ many K gates, so our algorithm is at most 2 times worse than the optimal K-count.

Clifford+CS circuit D is also an alternating sequence of generalized permutation part P_i and a K_1 gate (by replacing all appearences of K_0 by $Ex \cdot K_1 \cdot Ex$):

$$D = P_0 \cdot K_1 \cdot P_1 \cdot K_1 \cdot \ldots \cdot P_n. \tag{3}$$

So we have $\mathrm{lde}(D) \leq \mathrm{rkc}(D)$ and then $\mathrm{lde}(D) \leq \mathrm{kc}(D)$. On the other hand, Theorem IV.1 outputs D' such that $\mathrm{rkc}(D') \leq 2\mathrm{lde}(D)$, so we have $\mathrm{kc}(D) \leq 2\mathrm{lde}(D)$.

$$\mathrm{lde}(D) \leq \mathrm{kc}(D) \leq 2\mathrm{lde}(D)$$

If (3) is K-optimal, i.e., $\mathrm{kc}(D) = n$, then

$$\mathrm{rcs}(D) \leq \mathrm{kc}(D) + 1 \implies \mathrm{cs}(D) \leq \mathrm{kc}(D) + 1.$$

C. Lemmas and proofs

We start with a number of lemmas. Since $\mathbb{Z}[i]$ and $\mathbb{D}[i]$ are subrings of $\mathbb{C}$, the usual properties of complex numbers, complex vectors, and complex matrices, such as complex conjugation (denoted by $\bar{\cdot}$), inner product, norm, and conjugate transposition (denoted by $\cdot^\dagger$) are naturally inherited. For example, for a vector $v \in \mathbb{D}[i]^n$, the *norm* of v is $\|v\| = \sqrt{v^\dagger v}$, where $v^\dagger$ is the conjugate transpose of v. Note that $\|v\|^2 = v^\dagger v \in \mathbb{D}$, since $v^\dagger v \in \mathbb{D}[i] \cap \mathbb{R}$. Similarly if $v \in \mathbb{Z}[i]$, then $v^\dagger v \in \mathbb{Z}$. A *unit vector* is a vector of norm 1.

If v is a vector, the notation v_j refers to its jth entry; entries are numbered starting from $j = 0$.

Lemma IV.3. *For any $A \in U(\mathbb{D}[i])$ with lde k, there exists permutation matrix L and R such that, $\rho_1^k(LAR)$ or $\rho_1^k((LAR)^T)$ equals one of the following*

$$(i)\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix},\quad (ii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (iii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{bmatrix},$$

$$(iv)\begin{bmatrix}1&1&1&1\\1&1&1&1\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (v)\begin{bmatrix}1&1&0&0\\1&1&0&0\\1&1&1&1\\1&1&1&1\end{bmatrix},\quad (vi)\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}.$$

We also say A has such pattern and write $\mathrm{pat}(A)$ *for the pattern of A. Moreover, A has pattern (i) if and only if $\mathrm{lde}(A) = 0$, i.e., when A is a generalized permutation. A has pattern (iv), (v), and (vi) only if $\mathrm{lde}(A) > 1$.*

Before proving this lemma, we prove serveral useful lemmas.

Lemma IV.4. *Let v be a unit vector in $\mathbb{D}[i]^4$ with lde $k > 0$ then there exists a generalized unitary permutation matrix Q such that $\rho_1^k(Qv)$ equals $(1,1,0,0)^T$ or $(1,1,1,1)^T$. Moreover if $k > 1$ there exists a generalized unitary permutation matrix Q such that $\rho_2^k(Qv)$ equals to $(10,10,01,00)^T$ or $(10,10,10,10)^T$, and if $k = 1$, $\rho_2^1(Qv)$ equals to $(10,10,00,00)^T$.*

Proof. Let $w = \gamma^k v \in \mathbb{Z}[i]^4$. We first show w has an even number of odd entries. We have

$$\sum_{j=0}^{3} \|w_j\|^2 = \|w\| = |\gamma|^{2k}\|v\| = (\gamma\gamma^\dagger)^k \equiv 0 \,(\mathrm{mod}\ \gamma).$$

Therefore, there are an even number of $j \in \{0, ..., 3\}$ such that $\|w_j\|^2$ is odd. It follows by Lemma II.4 that an even number of w_j are odd.

We can find a permutation matrix P such that $\rho_2^k(Pv) = (1a, 1b, 0c, 0d)^T$ or $(1a, 1b, 1c, 1d)^T$. For the later, by muliplying a diagonal matrix, i.e., multiplying i to the required entries, we can make the second digit all zero, i.e., $(10, 10, 10, 10)^T$.

For the former, by muliplying a diagonal matrix, we can make second digit of the odd entries zero, i.e., $(10, 10, 0c, 0d)^T$. Now applying K to the pair of odd entries we get $(00, 01, 0c, 0d)^T$ or $(01, 00, 0c, 0d)^T$. Note that multiplication of v and a unitary is still a uinit vector, so is applying K to the pair of odd entries. So $(01, 00, 0c, 0d)^T$ or $(00, 01, 0c, 0d)^T$ is a residue of a unit vector with lde $k-1$, then its one-digit $k-1$ residue must of the form $(1, 1, 0, 0)^T$ upto a permutation, that is $c = 1$ and $d = 0$ upto a permutation.

If $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ upto a permutation. □

Lemma IV.5. *Let u, v be two unit vectors in $\mathbb{D}[i]^4$ with lde $k > 0$ such that $u \perp v$, i.e., $u^\dagger v = 0$. Then there exists a generalized unitary permutation matrix Q such that $\rho_1^k(Q[u|v])$ equals*

$$\begin{bmatrix}1&1\\1&1\\0&0\\0&0\end{bmatrix},\ \begin{bmatrix}1&0\\1&0\\0&1\\0&1\end{bmatrix},\ \begin{bmatrix}1&1\\1&1\\1&0\\1&0\end{bmatrix},\ \begin{bmatrix}1&1\\1&1\\0&1\\0&1\end{bmatrix},\ or\ \begin{bmatrix}1&1\\1&1\\1&1\\1&1\end{bmatrix}.$$

Proof. By Lemma IV.4, there can be only even number of 1s in each column. The only missing case is

$$\begin{bmatrix}1&0\\1&1\\0&1\\0&0\end{bmatrix}.$$

Then $\rho_1^k(u^\dagger v) = \sum_{j=0}^{3} \rho_1^k(\bar{u}_i v_i) = 1$ contradicts that $\rho_1^k(u^\dagger v) = \rho_1^k(0) = 0$. In other words, the number of indices such that at which the $(1, k)$ residue of u, v is 1 must be even. □

Proof of Lemma IV.3. Since the rows and columns of a unitary matrix are unit vectors and all row(column) vectors are perpendicular to each other. Lemma IV.4 and IV.5 forces the pattern to be (ii) - (vi) if $k > 0$. If $k = 0$, the only unitaries with lde 0 are the generalized permutations which has pattern (i). If $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ upto a permutation. So residue of such vectors cannot have four 1s. So Case (iv), (v), and (vi) only happens if the lde > 1. □

Lemma IV.6. *For any $A \in U(\mathbb{D}[i])$ with lde $k > 0$, there exist generalized permutations L, R, L', R' such that*

- *$\mathrm{lde}(K_1 LAR) \le k - 1$;*
- *or $\mathrm{lde}(LARK_1) \le k - 1$;*
- *or $\mathrm{lde}(L'K_1 LARR'K_1) \le k - 1$;*
- *or $\mathrm{lde}(K_1 L'K_1 LARR') \le k - 1$;*
- *or $\mathrm{lde}(L'CKLAR) \le k - 1$ (only happens when $\mathrm{lde}(A) = 1$).*

Note that each $\le$ above is actually $=$, but we don't need this. The proof proceeds by a case distinction over $\rho_1^k(A)$. Note that requiring $k > 0$ eliminates Case i. So start with Case ii.

1. Case ii

If $k > 1$, by Lemma IV.4, upto multilying generalized permutations on the left and right, $\mathrm{pat}(A)$ refines to

$$\begin{bmatrix}10&10&01&00\\10&1a&0b&0\bar{b}\\01&0c&0d&0e\\00&0\bar{c}&0f&0g\end{bmatrix},$$

where $\bar{b} = b \oplus 1$ and b, c, d, e, f, g are indeterminates. We claim $a = 0, b = c = 1$. Taking inner product of the first two cloumn gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Taking inner product of the second and third row gives

$$01 + 0c + 00 + 00 = 00 \Rightarrow c = 1.$$

Similarly, we have $b = 1$. Now pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 01 & 00 \\ 01 & 01 & 0d & 0e \\ 00 & 00 & 0f & 0g \end{bmatrix}.$$

Moreover, since $k > 1$ and the third and fourth row (column) is of lde $k - 1 > 0$, $d = f$, $d = e$, $e = g$, i.e., $d = e = f = g$. Now pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 01 & 00 \\ 01 & 01 & 0d & 0d \\ 00 & 00 & 0d & 0d \end{bmatrix}.$$

Applying $K_1 \cdot S_1$ on the left gives Case iv.

If $k = 1$, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 0d & 0e \\ 00 & 00 & 0f & 0g \end{bmatrix},$$

As before, we have $a = 0$. Apply $X_0 \cdot CK \cdot X_0$ on the left decreases the lde.

2. Case iii

If $k > 1$, by Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 0b & 0\bar{b} \\ 01 & 0c & 10 & 10 \\ 00 & 0\bar{c} & 10 & 1d \end{bmatrix},$$

Taking inner product of the first two cloumn gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Similarly $d = 0$. Taking inner product of the second and third row gives

$$01 + 0c + 0b + 0\bar{b} = 00 \Rightarrow c = 0.$$

Similary, we have $b = 0$. pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 01 & 00 & 10 & 10 \\ 00 & 01 & 10 & 10 \end{bmatrix}.$$

Applying $K_1 S_1$ on the left bring us to Case vi.

If $k = 1$. By Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 10 & 10 \\ 00 & 00 & 10 & 1d \end{bmatrix}.$$

As before, we have $a = 0, d = 0$. Applying K_1 on the left decrease the lde.

3. Case iv

This case only happens for $k > 1$. By Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1b & 1c \\ 01 & 0d & 0e & 0f \\ 00 & 0\bar{d} & 0\bar{e} & 0\bar{f} \end{bmatrix}.$$

Taking inner product of the first two cloumn gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Similarly we have $b = c = 0$. Since the third row is of lde $k - 1 > 0$, either $d = e = f = 1$ or one of d, e, f is 1 and the rest are 0. For the later, upto a permutation of columns, we may assume $d = 1, e = f = 0$. For both cases, right multiplication of K_1 decrease the lde.

4. Case v

This case only happens for $k > 1$. By Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 00 & 01 \\ 10 & 1b & 10 & 1f \\ 10 & 1c & 1e & 10 \end{bmatrix},$$

We claim $a = 0, b = c = e = 1$. Taking inner product of the first two row gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Taking inner product of the first and third row gives

$$10 + 1b + 01 + 00 = 00 \Rightarrow b = 1.$$

Taking inner product of the second and fourth row gives

$$10 + 1c + 00 + 01 = 00 \Rightarrow c = 1.$$

Taking inner product of the second and third column gives

$$01 + 00 + 11 + 1\bar{e} = 00 \Rightarrow e = 1.$$

Similarly $f = 1$. Now pattern is

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 10 & 11 & 10 & 11 \\ 10 & 11 & 11 & 10 \end{bmatrix},$$

Applying K_1 on the left bring us to Case iv.

5. *Case vi*

This case only happens for $k > 1$. By Lemma IV.4, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1d & 1g \\ 10 & 1b & 1e & 1h \\ 10 & 1c & 1f & 1x \end{bmatrix},$$

Taking inner product of first two columns

$$a + b + c = 0 \Rightarrow a, b, c \text{ cannot be all 1.}$$

Upto a permutation, we maybe assume $a = 0$. Then $b = c$. Similarly $d = g$.

If $b = c = 1$, and $d = g = 1$, pat(A) is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1h \\ 10 & 11 & 1f & 1x \end{bmatrix},$$

Taking inner product of the first and third column gives $f = \bar{e}$. Similarly, $h = \bar{e}$ and $x = \bar{f}$. Now pat(A) is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1\bar{e} \\ 10 & 11 & 1\bar{e} & 1e \end{bmatrix}.$$

We prove this pattern cannot happen by looking at its (3,k)-residue. Upto multiplication oof powers of -1 on the proper rows and columns, the (3,k)-residue refines to

$$\begin{bmatrix} 101 & 101 & 101 & 101 \\ 101 & 10x & 11u & 11a \\ 101 & 11y & 1ev & 1\bar{e}b \\ 101 & 11z & 1\bar{e}w & 1ec \end{bmatrix}.$$

Unitarity of A and the $k > 1$ requires that the any two rows (columns) are perperdicular to each other. In particular, by taking inner product of the first columns and the second, third column, we have

$$101 + 10x + 11y + 11z = 000 \implies x \oplus y \oplus z = 1$$
$$101 + 11u + 1ev + 1\bar{e}w = 000 \implies u \oplus v \oplus w = 1$$

Taking inner product of the second and thired column gives

$$101 + 11(u \oplus \bar{x}) + (1ev + 01e + 00\bar{y}) + (1\bar{e}w + 01\bar{e} + 00\bar{z})$$
$$= 000 \implies x \oplus y \oplus z \oplus u \oplus v \oplus w = 1, \text{ a contradiction.}$$

If $b = c = 0$, taking inner product of the first and third column gives $e = h$, similarly, $f = x$. So pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 1d & 1d \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

Taking inner product of the first and third row gives $d \oplus e \oplus f = 0$, then at leat one of d, e, f is 0, upto a permutation of rows, we may assume $d = 0$. Similar analysis applies if $d = g = 0$. So, for either case, upto permutations of rows and columns pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

Note that $e = f$, which is implied by taking inner product of the first and third column. So left multiplication of K_1 reduces the lde.

The transitions between cases in summarized in Figure 1.

$$\begin{tikzcd}
\text{ii} \begin{bmatrix} 1&1&0&0 \\ 1&1&0&0 \\ 0&0&0&0 \\ 0&0&0&0 \end{bmatrix} \arrow[r, "\text{if lde}=1,\ CK\times"] \arrow[d, "K_1\times"'] & \begin{bmatrix} 0&0&0&0 \\ 0&0&0&0 \\ 0&0&0&0 \\ 0&0&0&0 \end{bmatrix} & \text{iii} \begin{bmatrix} 1&1&0&0 \\ 1&1&0&0 \\ 0&0&1&1 \\ 0&0&1&1 \end{bmatrix} \arrow[l, "\text{if lde}=1,\ K_1\times"'] \arrow[d, "K_1\times"] \\
\text{iv} \begin{bmatrix} 1&1&1&1 \\ 1&1&1&1 \\ 0&0&0&0 \\ 0&0&0&0 \end{bmatrix} \arrow[ur, "\times K_1 \text{ or } K_1\times"] & \text{v} \begin{bmatrix} 1&1&0&0 \\ 1&1&0&0 \\ 1&1&1&1 \\ 1&1&1&1 \end{bmatrix} \arrow[l, "K_1\times"] & \text{vi} \begin{bmatrix} 1&1&1&1 \\ 1&1&1&1 \\ 1&1&1&1 \\ 1&1&1&1 \end{bmatrix} \arrow[ul, "K_1\times"]
\end{tikzcd}$$

FIG. 1. Transitions between cases. The $K_1\times$ ($\times K_1$) annotated arrow means if A is of the pattern at the tail, then K_1LAR ($LARK_1$) is of the pattern at the head for appropritate choosen generalized permutations L and R. Similarly, for other annotations. Note that CK decomposes to two K gates as in Equation 1.

D. Complexity

The time complexity is $O(\mathrm{lde}(U))$.

V. EXPERIMENT

We implemented our algorithm in Haskell and did an comparison with the CS-optimal synthesis algorithm. Circuits are randomly genrated feeded to both algorithm, and output circuits are compared on metrics $\mathrm{rcs}, \mathrm{rkc}, \mathrm{rlen}$.

Statistics show our algorithm alway acheives the optimal CS-count, which suggests that part of our analysis is not tight, and could be improved. This is left to futurn work.

VI. CONCLUSION AND FUTURE WORK

In conclusion, we proposed an efficient and optimal algorithm for the exact synthesis of Clifford+CS circuits from a given unitary matrix.

The algorithm Li et al. gave outputs a circuit that is at most 10 times worse than optimal in T-count. We believe our method can be used to improve the bound 10.

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